Ho Chi Minh City Bus GPS Dataset

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Abstract

This document describes a bus Global Positioning System (GPS) trajectory dataset from the Ho Chi Minh City, Vietnam public bus network. The dataset is prepared specifically for educational and research purposes. All sensitive and identifiable information has been anonymized and encoded to ensure data privacy and ethical use.

The dataset captures real-world, large-scale, streaming like GPS messages generated by buses operating in Ho Chi Minh City. It is designed to practice data ingestion, preprocessing, large-scale data analysis, and exploratory analytics using big data tools and frameworks.

Keywords: GPS, bus trajectory

1 Background: Ho Chi Minh City Bus Network

Ho Chi Minh City (HCMC) operates one of the largest urban bus networks in Vietnam, serving hundreds of thousands of passengers daily. The bus system is operated and coordinated under the supervision of the Ho Chi Minh City Management Centre of Public Transport (MCPT), which is responsible for planning, managing, and regulating public bus services across the city. The network consists of:

- A dense set of inner-city routes.
- Suburban and inter-district routes.

- Multiple types of terminals and stations (terminals, transfer points, regular stops).

The bus system plays a critical role in urban mobility and is currently being integrated with new public transport infrastructure, including the metro system. As a result, the bus network provides a rich and realistic data source for transportation analytics and big data research.

1.1 Bus Route Paths

In HCMC, it is common for a specific route number to encompass more than one physical path. We categorize these path variations into two scenarios, which are visually detailed in Figure 1. The initial scenario features two distinct paths: an outbound path and an inbound path, which link a designated start terminal with an end terminal. The second scenario involves only one primary path, which usually signifies a loop transit route.

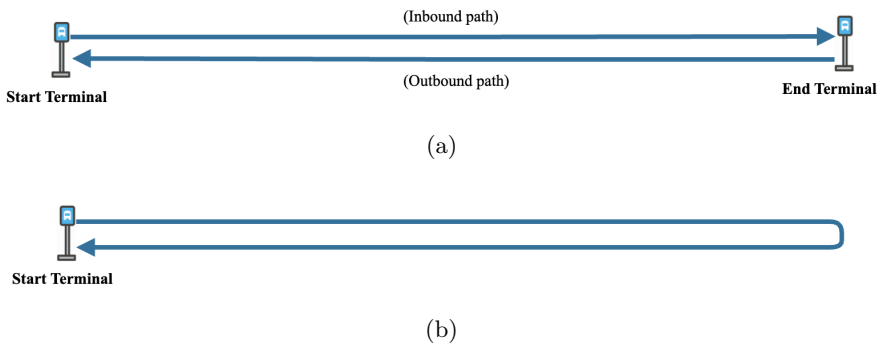

Fig. 1: Two primary route path cases in HCMC.

1.2 Terminal and Bus Stop Locations

The bus infrastructure in HCMC includes five variations of bus stops, where it is common for a single location to be utilized by multiple bus routes. An example of the structure of this bus stop dataset is presented in Table 1. This data structure captures critical attributes for each stop: a unique identifier, the stop name, its defined type, the street it is located on, its geographical coordinates (lng and lat), and a comprehensive field listing all route numbers (routes) that pass through this stop.

1.3 Route Timetable

The bus route timetable represents the proposed schedule for each transit line. Every bus service provider is required to submit and inform the MCPT of their planned schedule conditions dataset. These timetables specify expected operating conditions,

Table 1: Example terminal and bus stop locations.

Stop ID	Name	Type	Street	lng	lat	Routes
725	Ben xe Mien Tay	Ben xe	Kinh Duong Vuong	106.619438	10.739486	102, 151, 41
874	Chua Vien Minh	Tru dung	Nguyen Van Linh	106.660316	10.723442	102
7940	Cho Thanh My Loi	Nha cho	Pham Than Duat	106.767532	10.769271	99
1853	Sieu thi BigC	O son	Ho Hoc Lam	106.604934	10.732233	151, 61
1491	Cho Tam Ha	Bien treo	To Ngoc Van	106.747761	10.859586	141
...	...	...	...	...	...	...

such as trip start and end times, planned headways, and general frequency patterns under normal circumstances.

It is important to note that route timetables reflect planned operations rather than actual real-time performance. In practice, observed bus trajectories derived from GPS data may deviate from the published timetable due to traffic congestion, road closures, special events, weather conditions, or operational adjustments mandated by MCPT.

Noted: Detailed information about route paths, terminals, stations and official route timetable is not fully included in this dataset. You are encouraged to enrich the dataset by crawling or collecting additional information from the official website of MCPT (**buyttphcm.com.vn**), if needed, for extended analysis.

2 Selected Bus Routes

From the full HCMC bus network, **31 routes** were identified as representative samples for this dataset. These routes, listed in Table 2, reflect the network's inherent diversity in terms of length, duration, frequency, stop density and connected with Metro line 1. By selecting this subset, this dataset captures essential real-world operational dynamics while keeping the analytical complexity within practical limits.

3 Time Period of Data Collection

The dataset covers a specific and continuous **50-day** time window, from **March 20 to May 10, 2025**. This period was deliberately selected to capture a wide range of operational and traffic conditions in HCMC, allowing for robust temporal and comparative analysis. The selected time window includes:

- Regular working days, reflecting typical weekday traffic demand and commuting patterns.
- Weekends, which often exhibit different mobility behaviors, service frequencies, and congestion levels.
- Major public holidays and special events, including: Hung Kings Commemoration Day (07/04); Reunification Day (30/04); International Workers' Day (01/05); The April 30th military parade period (24/04–30/04).

During the April 30th military parade period, extensive road closures and traffic control measures were implemented in central areas of the city. As a result:

Table 2: Selected bus routes.

Route No	**Route Name**	**Duration** (min)	**Distance** (km)	**Inbound/Outbound stops**	**Connected to Metro 1**
70-5	Bo Heo - Loc Hung	15	6.00	2/2	✓
61-7	Ben do Binh My - Ben xe Binh Duong	15	6.60	2/2	
156D	Ben Thanh - Benh vien Mat	30	7.35	24	✓
156V	Ben Thanh - Ga Hoa Hung	30	7.35	23	✓
169D	Ga Binh Thai - Cang Phuc Long	25	7.90	22	✓
169V	Ga Binh Thai - Vincom Thu Duc	25	7.90	21	✓
163D	Ga Binh Thai - Cao dang Cong Thuong	25	8.00	24	✓
163V	Ga Binh Thai - Truong Phuoc Binh	25	8.00	25	✓
1	Ben Thanh - Ben xe Cho Lon	35	8.59	28/33	✓
164D	Dai hoc Nong LAm - Chung cu Topaz	35	10.40	19	✓
164V	Dai hoc Nong Lam - Ga Khu Cong nghe cao	35	10.40	20	✓
57	Cho Phuoc Binh - Truong Hiep Binh	42	13.10	34/33	
167D	Dai hoc Nong Lam - Ga Khu Cong nghe cao	55	13.15	34	✓
167V	Dai hoc Nong Lam - Khu che xuat Linh Trung	55	13.15	37	✓
152	Trung Son - San bay Tan Son Nhat	50 - 55	14.55	41/35	✓
27	Ben xe Sai Gon - Ben xe An Suong	55 - 65	15.60	42/38	
148	Ben xe Mien Tay - Go Vap	50 - 65	16.70	41/49	
55	Quang Trung - Ben xe Van Thanh	45	16.80	51/42	
151	Ben xe Mien Tay - Ben xe An Suong	50	17.20	39/35	
30	Cho Tan Huong - Ben xe Van Thanh	50	18.30	49/56	
122	Ben xe An Suong - Tan Quy	50	19.76	43/38	
3	Ben Thanh - Thanh Xuan	60 - 65	20.60	52/55	✓
72	Ben xe Sai Gon - Hiep Phuoc	55	21.60	42/41	
45	Dai hoc Kinh Te - Ben xe Mien Dong	70 - 80	22.30	56/58	✓
88	Ben Thanh - Cho Long Phuoc	65	23.60	55/53	✓
32	Ben xe Mien Tay - Ben xe Nga tu Ga	80	24.80	71/74	
91	Ben xe Mien Tay - Cho Thu Duc	80	25.10	67/65	
93	Ben Thanh - Ben xe Mien Dong moi	70 - 80	26.20	55/55	✓
24	Ben xe Mien Dong - Hoc Mon	75	27.20	67/64	
50	Dai hoc Bach Khoa - Dai hoc Quoc Gia	65	28.60	59/60	
90	Pha Binh Khanh - Can Thanh	75	45.60	29/28	

Table 3: Dataset feature descriptions.

Feature	Data type	Description
vehicle	string	The bus license plate.
driver	string	The driver's and license number.
speed	float	The current speed in km/h.
datetime	int32	The precise datetime, recorded as a Unix Time timestamp.
x	double	The latitude.
y	double	The longiute.
heading	float	Compass direction.
ignition	bool	True/False
aircon	bool	THe air conditioning status.
door_up	bool	The front door status.
door_down	bool	The rear door status.
sos	bool	SOS signal.
working	bool	School bus.

- Several bus routes were temporarily adjusted or detoured.
- Some routes experienced changes in their operating paths.
- In certain cases, terminal stations were relocated or modified for operational reasons.

These exceptional conditions introduce noticeable deviations from normal bus operations and should be treated as important events of interest rather than data noise. They provide valuable opportunities for analyzing:

- The impact of large-scale public events on public transport operations.
- Differences in travel time, trajectory patterns, and service reliability between normal days and disrupted periods.

4 Data Format and Structure

The dataset is stored in **JSON format**. Each record represents a GPS message similar to those sent directly from a bus to MCPT server in real time. Each JSON object encompasses several critical attributes, the specifications of which are provided in Table 3. Example of the GPS trajectory entries for bus H100 are presented in Table 4.

Table 4: Example GPS data of a bus with the license plate number H100.

vehicle	datetime	lat	lng	speed	door_up	door_down	driver
H100	2025-03-24 05:00:05	10.843127	106.612823	4.0	False	False	HPCLab
H100	2025-03-24 05:00:15	10.843102	106.612900	6.0	False	False	HPCLab
H100	2025-03-24 05:00:25	10.843198	106.612961	2.0	False	False	HPCLab
H100	2025-03-24 05:00:35	10.843227	106.612984	1.0	True	True	HPCLab
H100	2025-03-24 05:00:45	10.843246	106.612991	1.0	False	False	HPCLab
...	...	...	...	...	...	...	...

Anonymization and data privacy: To protect privacy and sensitive operational information: bus license plate and driver's name, license number have been anonymized and no personal data is included. The dataset is intended strictly for academic purposes.

5 Dataset Size and Storage

The complete dataset is hosted on **Kaggle** and on the **HPC Lab shared file system** (sharefile.hpcc.vn) (*links will be provided separately*). After decompression, the dataset occupies approximately **34GB**, reflecting the high temporal resolution and large volume of GPS messages collected over the 50-day period, necessitating the use of high-performance data processing techniques such as partitioning, sampling, or distributed computing frameworks.

To facilitate initial schema verification and exploratory analysis, a representative sample dataset is provided. This sample file mirrors the structural integrity of the primary dataset while remaining computationally lightweight for rapid inspection.

6 Dataset Highlights and Special Notes

This dataset contains several characteristics that distinguish it from simplified or synthetic transportation datasets. You should carefully consider the following highlights and special conditions when conducting analysis:

- Public holidays and special events: Travel behavior, traffic conditions, and bus operating patterns during public holidays differ significantly from regular weekdays. Reduced demand, altered service frequencies, and atypical congestion patterns may be observed.
- Large-scale event disruptions: Periods such as the April 30th military parade introduce non-recurring disruptions, including road closures, traffic control measures, and temporary rerouting. These events create clear structural breaks in trajectory patterns and travel times and should be explicitly modeled rather than treated as noise.
- Irregular sampling and missing data: As with most real-world GPS-based datasets, sampling intervals may be inconsistent, and certain records may be missing due to communication loss or operational issues. Robust preprocessing and data quality assessment are therefore essential.
- Route-specific operational behaviors: Different bus routes exhibit distinct characteristics depending on their geographic coverage, service type, and traffic environment. Inner-city routes, suburban routes, and long-distance routes may show markedly different speed profiles, dwell times, and reliability levels.
- Bus routes connected to metro lines: Selected routes serve as feeders or connectors to metro stations. These routes may demonstrate unique temporal patterns, peak-hour behavior, and passenger demand dynamics associated with intermodal transfers.

Taken together, these characteristics make the dataset particularly suitable for advanced big data tasks, including anomaly detection, spatio-temporal pattern mining, travel time modeling under non-stationary conditions, and comparative analysis between normal and disrupted periods.

7 Intended Use

This dataset is primarily intended for academic research in transportation analytics and urban mobility in the context of HCMC, with a strong emphasis on large-scale data-driven methods. It is designed to support both methodological development and empirical analysis using real-world GPS trajectory data derived from the HCMC public bus network, which operates under complex and dynamic urban traffic conditions.